

**DHARESHWAR VIDYA VA KARIDA
PRATISHTHAN'S**

**DHARASHWAR SHIKSHAN SHASTRA
MAHAVIDYALAYA,
DHAYARI, PUNE**

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1. INTRODUCTION:

In today's modern and technology-driven era, Information and Communication Technology (ICT) is widely used in the field of education. Due to the advancement of science and technology, significant changes have taken place in traditional teaching methods, and education has become more modern, accessible, and student-centered. With the help of tools such as computers, the internet, smartphones, digital boards, projectors, e-learning platforms, and educational apps, the teaching-learning process has become more effective and engaging.

The use of ICT enables teachers to make teaching clearer, more visual, and easier to understand by using various digital tools. The use of images, videos, animations, presentations, online quizzes, and digital notes increases students' interest in learning and helps them understand concepts more effectively. As a result, teaching becomes more interactive and student-centered.

Digital education provides students with opportunities for self-learning. They can access online materials, video lectures, e-books, notes, and question banks anytime and anywhere. This has significantly reduced the limitations of time and distance in education. Students from rural and remote areas are also gaining easier access to quality education.

Moreover, with the help of ICT, teachers can carry out tasks such as student assessment, attendance, results, and progress reports more quickly and accurately. Online examinations, digital assignments, virtual classrooms, and webinars have made education more modern and effective.

In today's era of global competition, it has become essential to develop digital skills among students. ICT-based education helps enhance students' creativity, problem-solving ability, research attitude, and technical skills. Therefore, ICT holds great importance in the modern education system, and digital education has become the need of the hour.

2. IMPORTANCE OF ICT AND DIGITAL EDUCATION

The use of Information and Communication Technology (ICT) in the field of education makes the teaching-learning process more modern, simple, and effective. Digital education has changed the way knowledge is acquired, and students have become more actively involved in learning. The importance of ICT in education is as follows:

✓ 1. Teaching becomes more attractive and effective

With the help of ICT, teachers can explain topics using videos, animations, presentations, smart boards, images, and digital models. The use of audio-visual media helps students stay focused and understand the subject better. This increases their interest in learning.

✓ 2. Information becomes easily and quickly available

Updated information on various subjects, e-books, research articles, educational videos, notes, and question banks are easily available on the internet. Students and teachers can search for necessary information anytime. This makes the process of gaining knowledge easier and faster.

✓ 3. Encourages self-learning

Digital education allows students to study independently through online courses, video lectures, educational apps, and websites. They can learn at their own pace, revise topics, and clarify their doubts. This increases students' confidence and self-reliance.

✓ 4. Saves time and effort

Digital notes, PPTs, online assignments, emails, virtual classes, and online examinations help save time for both teachers and students. The need for paperwork, notebooks, and record-keeping is reduced. This makes education more organized and convenient.

✓ 5. Enhances teachers' efficiency

With the help of ICT, teachers can easily prepare lesson plans, digital worksheets, question papers, presentations, assessments, and progress

reports. Online tools enable quick and accurate evaluation of students. This makes teachers' work easier and more effective.

✓ **6. Enables inclusive education**

ICT is highly useful for students with special needs. Audio books, screen readers, large fonts, video lectures, and interactive software allow all students to learn according to their abilities. Thus, education becomes equal and accessible for everyone.

✓ **7. Digital skills are developed among students**

Through the use of ICT, students gain knowledge of computers, the internet, presentations, online research, and digital communication. These skills are extremely essential for future education and employment.

✓ **8. Connects learners to global education**

Digital education enables students to access worldwide knowledge, online courses, webinars, and guidance from experts. This broadens their knowledge and helps develop a global perspective.

Information and Communication Technology has made education more modern, accessible, inclusive, and effective. Therefore, ICT-based education is extremely essential for quality education in today's era.

3. NEED AND BENEFITS OF ICT AND DIGITAL EDUCATION:

In today's information-technology era, major changes are taking place in the field of education. Along with traditional teaching methods, the use of digital tools is increasing. As a result, education has become more modern, effective, and inclusive. The need for ICT and the benefits of digital education can be explained as follows:

Need of ICT in Education

✓ 1. Essential for modern education

In today's technology-driven world, computers, the internet, mobile phones, and smart boards are widely used. It has become necessary for students to acquire digital skills. Knowledge of ICT is extremely important for future education and employment.

✓ 2. Growth of online education

During the COVID-19 period, the importance of online education was strongly highlighted. Online classes, digital examinations, webinars, and e-learning platforms have increased the need for ICT. Schools, colleges, and universities are widely adopting online education.

✓ 3. Need for quality education

Through ICT, students in rural and remote areas can also access quality education. Expert video lectures, e-books, and online courses are open to all. This helps reduce inequality in education.

✓ 4. Quick dissemination of information

Teachers can instantly send notes, homework, instructions, and study materials to students through email, WhatsApp, or learning platforms. This makes communication faster and easier.

✓ 5. Student-centered learning

ICT makes education more interactive. Students actively participate through projects, presentations, and online research. Thus, the learning process becomes student-centered.

✓ **6. Necessary for effective evaluation**

Online tests, digital quizzes, and automatic evaluation make assessment more accurate and faster. Teachers receive immediate results and analysis.

Benefits of Digital Learning:

✓ **1. Anywhere & Anytime Learning**

Digital education allows students to study anytime and anywhere. There are no limitations of time and place, making learning more flexible.

✓ **2. Cost-effective education**

E-books, digital notes, and online courses are available at low cost or free of charge. This reduces students' expenses. The use of paper, printing, and notebooks is also minimized.

✓ **3. Updated information**

Information on the internet is continuously updated. Students and teachers receive the latest knowledge, keeping education current and relevant.

✓ **4. Interactive learning**

Online quizzes, videos, games, animations, and simulations make learning enjoyable and interesting. Students learn with greater enthusiasm.

✓ **5. Environment-friendly education**

Digital tools reduce the use of paper. This helps in conserving trees and protecting the environment. Therefore, digital education is eco-friendly.

✓ **6. Increases self-reliance and confidence**

Students conduct research, gather information, and give presentations independently. This enhances their confidence and decision-making ability.

✓ **7. Saves time**

Digital notes, automatic evaluation, and online assignments save time. Both teachers and students can complete tasks more quickly.

✓ **8. Connection with global knowledge**

Digital education enables students to access courses, research, and expert guidance from around the world. This broadens their perspective.

ICT and digital education have become an integral part of the modern education system. ICT is extremely important for providing quality, inclusive, and effective education. Digital learning ensures the overall development of students and prepares them for the future. Therefore, ICT-based education is an essential need of the present time.

ACTIVITY 1

CREATING A DIGITAL NEWSLETTER:

1. Concept of a Digital Newsletter

A digital newsletter is a regular bulletin prepared by an institution, school, college, company, or group and delivered to readers through the internet. In today's digital era, various online tools are used to spread information quickly and effectively. One of the most important tools among them is the digital newsletter. Instead of traditional printed newsletters, a digital newsletter is sent through email, websites, or social media platforms.

Concept:

A digital newsletter is a planned, regular method of sharing information through digital media.

It generally includes the following:

- ✓ News and activities of the institution
- ✓ Educational articles and information
- ✓ Notices and announcements
- ✓ Information about upcoming events
- ✓ Students' achievements and success stories

The main objective of a digital newsletter is to maintain continuous communication with readers and provide them with updated information.

Features of a Digital Newsletter:

1. **Digital format** – Distributed through email, PDF, websites, or social media.
2. **Regular publication** – Published weekly, monthly, or quarterly.
3. **Attractive design** – Use of images, graphics, video links, and hyperlinks.
4. **Cost-effective** – No printing cost involved.
5. **Quick distribution** – Can be sent to many people with a single click.

6. **Interactive** – Readers can give feedback or responses.

Use in the Educational Field:

- ✓ To provide information about school and college activities
- ✓ To communicate with parents
- ✓ To publish students' achievements
- ✓ To share study materials and notices
- ✓ To announce competitions, events, and results

Benefits:

- ✓ Saves time and cost
- ✓ Environment-friendly (Paperless)
- ✓ Quick dissemination of information
- ✓ Attractive and modern presentation
- ✓ Reaches a wider audience

A digital newsletter is an effective tool for modern information sharing. It is widely used in education, business, and social sectors. Digital newsletters created with the help of ICT make information management simple and effective.

Therefore, in the digital era, a digital newsletter has become a highly useful medium for communication and information exchange.

2. Top 10 Educational Websites:

Website Name	Useful For
1. DIKSHA	E-learning materials for all classes and QR code-based textbooks.
2. e-Pathshala	Free collection of audio, video, and e-books.
3. SWAYAM	One of the largest online course platforms in the world.
4. Khan Academy	Concept-based videos for Mathematics and Science.
5. NDL (National Digital Library)	A vast collection of millions of books and resources.
6. Wikipedia	To get basic information on any subject.
7. PhET	Virtual experiments for Science and Mathematics.
8. GeoGebra	To simplify Geometry and Algebra.
9. British Council	To improve English language and grammar skills.
10. TED-Ed	Inspirational and educational videos for students.

3. Brief Description of Websites:

1. DIKSHA

Information: It is a digital education portal of the Government of India. E-books, videos, training materials, and lesson plans are available here.

Use: Teachers can use teaching materials and show additional videos to students.

2. SWAYAM

Information: A national platform that offers free online courses.

Use: Teachers and students can study new subjects and earn certificates.

3. NCERT

Information: A national educational body that provides official textbooks and curriculum.

Use: Teachers can access the syllabus and guidelines.

4. e-Pathshala

Information: All NCERT books are available in digital format.

Use: Students can read textbooks on their mobile devices.

5. NPTEL

Information: An online education program run by IITs and IISc.

Use: Provides video lectures and reference materials for higher education.

6. Khan Academy

Information: Offers free videos and practice exercises for Mathematics, Science, and other subjects.

Use: Helpful for students' self-study.

7. Coursera

Information: Provides online courses from universities around the world.

Use: Useful for teachers and learners to develop new skills.

8. BYJU'S

Information: A video-based digital learning platform.

Use: Helps students understand subjects in a simple way.

9. Vedantu

Information: Offers live online classes and study materials.

Use: Students can clarify doubts and revise topics.

10. Unacademy

Information: Provides courses for competitive exams and school subjects.

Use: Helps students with additional preparation.

11. Google Scholar

Information: Used to search for research papers and academic information.

Use: Teachers can access research materials.

12. Wikipedia

Information: A free online encyclopedia providing information on various subjects.

Use: Students can obtain basic information (should be verified before use).

All the above educational websites are highly useful for teachers and students. These platforms make digital learning more effective. Teachers get easy access to various digital teaching tools, and students get opportunities for self-learning.

4. Structure of Digital Newsletter:

Open Educational Resources (OER) and Modern Teaching

Education: From books to the digital revolution



Gyan Ganga: Now at your click!

In today's technological age, education is no longer limited to books but has become digital. 'Open Educational Resources' (OER) is a boon for teachers to access quality and free information.

Major educational websites Before using a website, make sure it is valid (eg .gov, .edu, etc.).



Education beyond boundaries

- **Free Availability:** Information is available without any charge.
- **Legal Use:** Information can be used and modified.
- **Quality Material:** Educational material created by the company.

Major Educational Websites



WEBSITES names	What is it useful for?
DIKSHA	DIKSHA E-learning materials and QR code textbooks for all classes.
e-Pathshala	Free collection of audio, video, and e-books.
SWAYAM	One of the largest online course platforms in the world.
Khan Academy	Concept-based videos for Mathematics and Science.
NDL (National Digital Library)	National Digital Library – a treasure of crores (millions) of books.
Wikipedia	For obtaining basic information on any topic.
PhET	Virtual experiments in Science and Mathematics.
GeoGebra	For making Geometry and Algebra easy.
British Council	To improve English language and grammar.
TED-Ed	Inspirational educational videos for students.

Good Thought : Technology will not replace teachers, but it will create teachers who use technology.

ACTIVITY 2

DIGITAL ASSESSMENT TOOL – ONLINE QUIZ

1) Concept of Digital Assessment

Digital assessment refers to evaluating students' knowledge, skills, and understanding through digital tools.

In this method, computers, mobile phones, the internet, and various online platforms (such as Google Forms, Kahoot!, Quizizz, etc.) are used.

In digital assessment, students are given online question papers, and their answers are checked instantly. This makes the evaluation process faster, more accurate, and transparent.

Features of Digital Assessment:

- ✓ Instant results
- ✓ Automatic evaluation
- ✓ Saves time
- ✓ Immediate feedback to students
- ✓ Can be used both online and offline

Google Form Link:

https://docs.google.com/forms/d/e/1FAIpQLSfNYZ7q_4DuWWFZ3z1lC1bom4II2UCGPuSrQkg1YXRGp8LEQ/viewform?usp=header

2) Structure of the Question Paper

Tool: Google Forms

Subject: Science

Unit/Topic: Structure of flower

Total Questions: 15 (Multiple Choice Questions – MCQs)

1) What is the main function of a flower?

- a) To prepare food
- b) To absorb water
- c) To help in reproduction
- d) To store food

2) Which part of the flower attracts insects and birds for pollination?

- a) Sepals
- b) Petals
- c) Ovary
- d) anther

3) The stamen consists of which two parts?

- a) Stigma and style
- b) Ovary and ovules
- c) Anther and filament
- d) Ovary and style

4) What is the primary function of sepals?

- a) To attract pollinators
- b) To protect the flower bud
- c) To form seeds
- d) To form fruit

5) After fertilization, the ovary develops into a

- a) Leaf
- b) Root
- c) Stem
- d) Fruit

6) Flowers help in the formation of

- a) Leaves
- b) Seeds
- c) Branches
- d) Roots

7) Which process helps transfer pollen from anther to stigma?

- a) Germination
- b) Respiration
- c) Pollination
- d) Photosynthesis

8) What attracts insects to the flower?

- a) Roots
- b) Stem
- c) Bright petals and nectar
- d) Soil

9) Flowers play an important role in:

- a) Digestion
- b) Transportation
- c) Reproduction of plants
- d) Breathing

10) The sweet liquid that attracts insects is called:

- a) Pollen
- b) Nectar
- c) Water
- d) Sap

11) Which of the following is NOT a function of a flower?

- a) Seed formation
- b) Fruit formation
- c) Pollination
- d) Photosynthesis

12) Fertilization in flowers leads to the formation of:

- a) Leaves
- b) Seeds
- c) Roots
- d) Soil

13) The reproductive organ of a plant is the:

- a) Leaf
- b) Stem
- c) Root
- d) Flower

14) Pollination helps in:

- a) Water absorption
- b) Food storage
- c) Reproduction
- d) Breathing

15) After fertilization, ovules develop into:

- a) Leaves
- b) Seeds
- c) Petals

3. Answer Sheet:

Question No	Answer
1	c) To help in reproduction
2	b) Petals
3	c) Anther and filament
4	b) To protect the flower bud
5	d) Fruit
6	b) Seeds
7	c) Pollination
8	c) Bright petals and nectar
9	c) Reproduction of plants
10	b) Nectar
11	d) Photosynthesis
12	b) Seeds
13	d) Flower
14	c) Reproduction
15	b) Seeds

4) Feedback Mechanism:

In a digital quiz, a feedback mechanism is extremely important. Students receive immediate information about their performance.

Types of Feedback:

- ✓ **Instant Feedback** – The correct/incorrect answer is shown after each question.
- ✓ **Score Display** – Total marks and percentage are displayed.
- ✓ **Correct Answer Explanation** – The correct answer is given along with an explanation.
- ✓ **Performance Analysis** – It shows which questions were answered incorrectly.
- ✓ **Encouragement Message** – Messages such as “Good Job,” “Try Again,” and “Keep Practicing” are displayed.

Benefits of Feedback:

- ✓ **Students immediately become aware of their mistakes.**
- ✓ **It helps in making improvements.**
- ✓ **It increases confidence.**
- ✓ **It motivates learning.**

ACTIVITY 3

DIGITAL SAFETY AWARENESS POSTER

1) Concept of Cyber Safety

Cyber safety means using the internet, computers, mobile phones, tablets, and other digital devices in a safe, responsible, and aware manner. Protecting oneself and one's personal information from potential online threats, fraud, and misuse is called cyber safety.

In today's digital age, students, teachers, and citizens widely use the internet. Online education, social media, online banking, email, e-commerce, gaming, and other services have become a part of daily life. However, along with these facilities, certain risks have also emerged.

Cyber Threats

In the digital world, the following types of threats may occur:

- ✓ Hacking – Unauthorized access to computers or accounts.
- ✓ Cyber Bullying – Harassing or sending insulting messages on social media.
- ✓ Identity Theft – Misuse of personal information.
- ✓ Phishing – Fraud through fake emails or links.
- ✓ Malware – Inserting harmful software into a computer.
- ✓ Online Fraud – Financial fraud in online transactions.

Need for Cyber Safety

Today, students start using mobile phones and the internet at an early age. Therefore, proper guidance is necessary. With awareness of cyber safety, students can stay away from harmful websites, unknown persons, and fraudulent messages.

Cyber safety helps in:

- ✓ Keeping personal information secure
- ✓ Making financial transactions safe
- ✓ Preventing mental harassment
- ✓ Using digital devices responsibly

Importance in the Educational Field

The use of digital tools is increasing in schools and colleges. Online classes, e-learning platforms, and digital assignments have increased students' interaction with the internet. Therefore, it is essential for educational institutions to create awareness about cyber safety.

Teachers should educate students about the following:

- ✓ Creating strong passwords
- ✓ Not sharing personal information
- ✓ Avoiding suspicious links
- ✓ Reporting cyber crimes

Responsible Digital Citizenship

Cyber safety is not only a technical issue but also a social responsibility. While using the internet, everyone must follow ethics, maintain proper behavior, and obey the law.

“Think Before You Click” is the key principle of cyber safety.

Cyber safety is extremely important for modern education and society. By using digital tools properly, safely, and responsibly, we can protect ourselves and society from cyber crimes. Creating awareness about cyber safety among students from an early age is the need of the hour.

2) Selected Topic: Online Safety

The selected topic for this activity is Online Safety.

Online Safety means protecting one's personal information, privacy, financial transactions, and digital identity while using the internet, social media, email, online games, digital payment apps, and e-learning platforms.

In the digital age, the internet has become a major medium for education, communication, entertainment, and transactions. Students use the internet for online classes, project research, video lectures, social networking, and gaming. However, not all information and people on the internet are safe. Due to a lack of proper guidance, students may fall victim to fraud, cyberbullying, hacking, phishing, and identity theft.

Therefore, creating awareness about Online Safety is the need of the hour.

Importance of Online Safety

- ✓ Protection of Personal Information – Information such as name, address, mobile number, passwords, and ID numbers remains secure.
- ✓ Financial Security – Online banking, UPI, and card transactions remain safe.
- ✓ Protection of Mental Health – Cyberbullying and online harassment can be prevented.
- ✓ Protection of Digital Identity – Prevents social media accounts from being hacked.
- ✓ Legal Awareness – Provides knowledge of cyber laws and helps prevent crimes.
- ✓ Development of Responsible Digital Citizens – Students use the internet ethically and responsibly.

Important Aspects of Online Safety

1) Password Safety

- Use a strong password with at least 8–12 characters.
- Include numbers, uppercase and lowercase letters, and special symbols.
- Do not use the same password everywhere.
- Change passwords regularly.

2) Privacy Settings

- **Keep social media accounts private.**
- **Adjust settings so that personal photos and information are visible only to trusted people.**
- **Turn off location sharing.**

3) Safe Browsing

- **Use websites that begin with “https://”.**
- **Do not click on suspicious pop-ups or advertisements.**
- **Download only official and trusted apps.**

4) Cyber Etiquette (Digital Etiquette)

- **Use polite language online.**
- **Respect others.**
- **Do not share false or misleading information.**
- **Follow copyright rules.**

5) Reporting System

- **Inform parents or teachers immediately about suspicious emails, links, or calls.**
- **Report cyber crimes to the concerned authorities.**
- **Use the “Report” option on social media platforms.**

6) Device Security

- **Use antivirus software on mobile phones and computers.**
- **Update software regularly.**
- **Be cautious while using public Wi-Fi.**

7) Screen Time Management

- **Avoid excessive internet use.**
- **Maintain a balance between study and entertainment.**
- **Plan time properly to prevent digital addiction.**

Online Safety and the Educational Field

As digital learning is increasing in schools and colleges, training on Online Safety is extremely important. Teaching students safe digital habits will help them become responsible citizens in the future.

3) Key Messages

✓ 1. Use Strong Passwords

Create passwords with at least 8–12 characters, including numbers and special symbols.

Example: Use a secure password like *Abc@1234*.

✓ 2. Do Not Share OTP or Passwords

OTP and passwords are confidential information. Never share them with anyone.

✓ 3. Do Not Click on Unknown Links

Avoid clicking on fake offers, lucky draw links, or prize-winning messages.

✓ 4. Do Not Share Personal Information on Social Media

Do not make your home address, school name, phone number, or bank details public.

✓ 5. Block Suspicious Messages or Calls Immediately

Block fraudulent calls and messages and report them.

✓ 6. Inform Parents/Teachers About Cyberbullying

If you face online harassment, do not stay silent—inform elders immediately.

✓ 7. Use Safe and Trusted Websites Only

Use official websites that begin with “https://”.

✓ 8. Follow Digital Etiquette

Use polite language on the internet and respect others.

✓ 9. Update Software Regularly

Keep mobile and computer apps, as well as antivirus software, updated.

✓ 10. Remember the Motto “Think Before You Click”

Think carefully before sharing any information or clicking on any link.

ACTIVITY 4

ANALYSING A MOOC

1) MOOC Concept:

MOOC stands for **Massive Open Online Course**.

A MOOC is an online course designed for large-scale participation and open access through the internet. It allows learners from different geographical locations to enroll in a course without any restriction of age, qualification, or location.

The concept of MOOCs emerged to promote **accessible, flexible, and affordable education** to a large number of learners simultaneously. MOOCs are usually offered by universities, educational institutions, and online learning platforms.

Importance of MOOCs in Education:

MOOCs support flexible learning, self-paced study, and blended learning approaches. They help teachers and students access quality educational content from reputed institutions without geographical limitations.

In teacher education, MOOCs help future teachers develop digital competency and enhance subject knowledge in alignment with modern educational needs.

2) Information about the Selected MOOC Platform – **SWAYAM:**

SWAYAM is a government of India initiative and a national level MOOC platform designed to provide **free and high-quality online education** to learners from Class 9 up to post-graduation and beyond. It was launched by the Ministry of Education under the Government of India with the aim of achieving the three cardinal principles of education policy — **access, equity, and quality**.

The platform allows anyone to enroll in courses from anywhere, at any time, without fee barriers, helping bridge the **digital divide** for learners who may not have access to traditional education. It provides options for learners to earn

certificate.SWAYAM's mission is to make educational resources available to underserved learners and to support lifelong learning opportunities.Courses on SWAYAM are developed and delivered

The courses follow a structured **4-Quadrant Learning Model**, which ensures a comprehensive online learning experience. Each course typically includes:

- **Video Lectures** created by subject experts
- **Downloadable Reading Material** for deeper understanding
- **Self-Assessment Tests and Quizzes** to evaluate learning
- **Online Discussion Forums** for doubt clearing and peer interaction

3) Course Objectives – Animal Biotechnology (SWAYAM):

The course Animal Biotechnology aims to provide comprehensive knowledge of modern biotechnological techniques applied to animals.

✓ Objectives of the Course:

- To introduce fundamental concepts of animal biotechnology and molecular biology.
- To familiarize learners with genome sequencing and gene expression analysis.
- To provide understanding of advanced genome editing technologies such as:
 - i. Zinc Finger Nucleases (ZFNs)
 - ii. TALENs
- To explain how genetic engineering can improve:
 - i. Disease resistance
 - ii. Pharmaceutical applications
- To develop theoretical and practical knowledge related to biotechnology techniques used in animal sciences.
- To prepare students for professional careers and higher studies in biotechnology and related fields.

Course Design – Animal Biotechnology (SWAYAM)

The Animal Biotechnology course is structured as a **12-week undergraduate elective course** with 4 credit points. The course is designed following the standard SWAYAM four-quadrant learning model.

1. Duration and Level

Course Level: Undergraduate

Duration: 12 Weeks

Credits: 4

Category: Zoology

2. Teaching-Learning Structure

The course is designed with the following components:

a) Video Lectures

Expert faculty provide recorded video lectures explaining theoretical concepts in a structured and progressive manner.

b) Study Materials

Downloadable reading materials and reference notes are provided to enhance conceptual understanding.

c) Weekly Assignments

Online quizzes and assignments are conducted to assess learners' understanding of each module.

d) Discussion Forum

An interactive discussion forum allows learners to clarify doubts and engage in academic discussions.

e) Final Examination

A proctored final examination is conducted after course completion. Successful candidates receive certification based on performance.

3. Pedagogical Approach

The course follows a blended digital pedagogy that includes:

- Concept-based teaching

- Case studies and applications

- Practical orientation

- Self-paced learning

- Continuous internal assessment

4. Analytical Report of the Selected MOOC Platform

In the digital era, **Massive Open Online Courses (MOOCs)** have emerged as an important and innovative concept in the field of education. MOOCs provide large numbers of learners with quality, flexible, and affordable education through the internet. For this analytical report, the Government of India's national MOOC platform **SWAYAM (Study Webs of Active Learning for Young Aspiring Minds)** and the course **Animal Biotechnology** have been selected.

Information about the Selected MOOC Platform – SWAYAM

SWAYAM is an initiative of the Ministry of Education, Government of India. The platform was developed to achieve the core principles of **Access, Equity, and Quality** in education. It offers free courses in various subjects from Class 9 to postgraduate level.

SWAYAM is based on the **4-Quadrant Learning Model**, which includes the following components in each course:

1. Video lectures by subject experts
2. Downloadable study materials
3. Self-assessment tests and weekly assignments
4. Online discussion forums for doubt resolution

This structure provides learners with a comprehensive and well-organized online learning experience.

Selected Course – Animal Biotechnology

“Animal Biotechnology” is a 12-week undergraduate-level course carrying 4 credits. The objective of this course is to provide in-depth knowledge of modern technologies in animal biotechnology.

Major Objectives of the Course:

- To explain fundamental concepts of animal biotechnology and molecular biology
- To understand genome sequencing and gene expression analysis
- To provide knowledge of advanced gene-editing technologies such as ZFNs and TALENs
- To explain applications of genetic engineering in disease resistance and pharmaceutical production
- To prepare students for higher education and professional careers

Course Structure and Teaching Methodology

The course includes weekly modules, video lectures, study materials, online assignments, and a final examination. It is based on concept-oriented teaching, an application-based approach, and self-paced learning. The discussion forum encourages interaction and collaborative learning among students.

Educational Significance

The SWAYAM platform helps students and teachers develop digital skills. From a teacher education perspective, it enhances subject knowledge, technical proficiency, and self-learning habits. The continuous assessment system makes the learning process more effective.

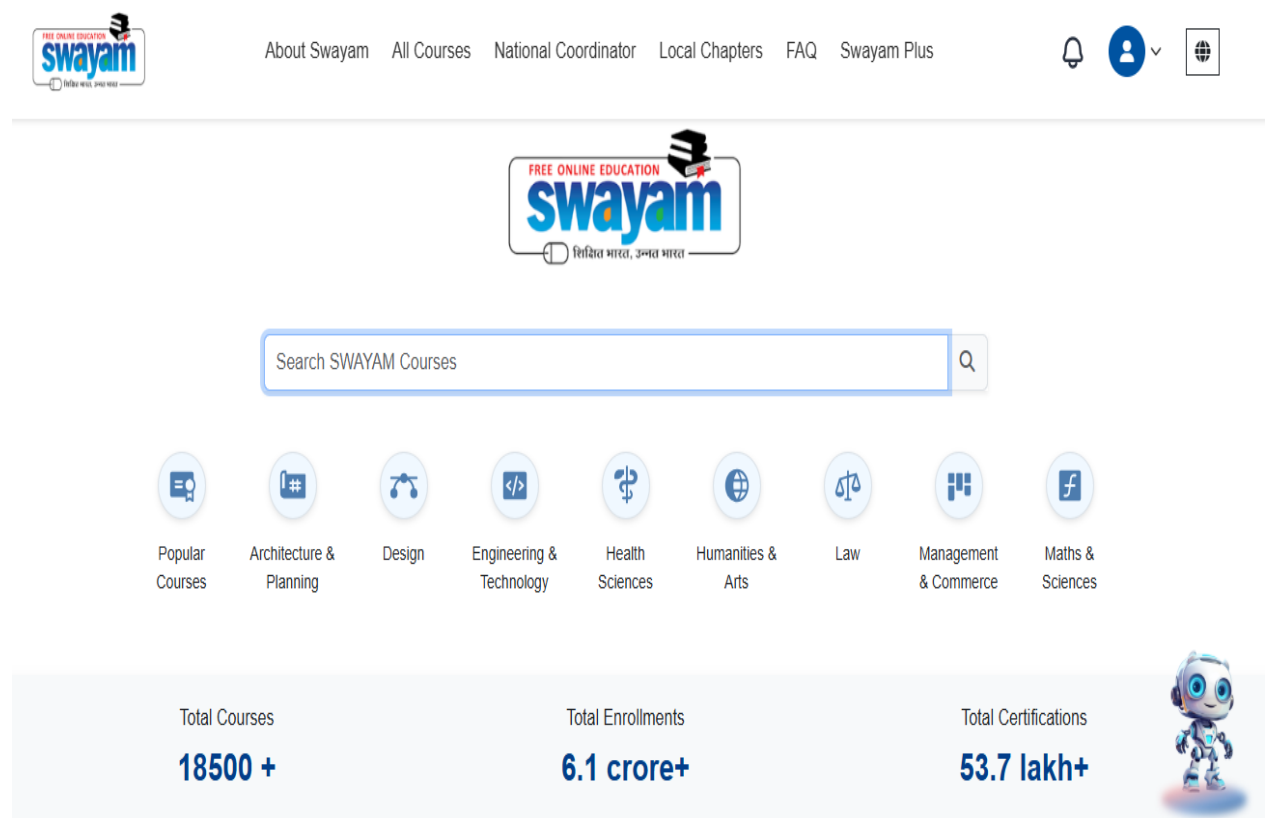
Conclusion

The analysis clearly shows that SWAYAM is a comprehensive, flexible, and high-quality MOOC platform. The course “Animal Biotechnology” is well-structured, career-oriented, and provides modern scientific knowledge. In the context of the growing demand for digital education, such MOOC platforms play a vital role in strengthening the future education system.

5.Screenshots:

Step 1: Opened the SWAYAM Platform

I opened the official **SWAYAM** website in a web browser by typing www.swayam.gov.in.



Step 2: Explored the Course Categories

On the homepage, I browsed through the available course categories and selected the relevant subject area.

The screenshot shows the Swayam homepage with a navigation bar at the top. The navigation bar includes the Swayam logo, links for 'About Swayam', 'All Courses', 'National Coordinator', 'Local Chapters', 'FAQ', 'Swayam Plus', and a user profile dropdown for 'rohinipatil616@g...'. Below the navigation bar, there are three course cards displayed in a grid. Each card shows the course title, instructor name, institution, and enrollment/exam dates. The first card is 'Animal Biotechnology' by Dr. Jayasree.S at Mercy College, Palakkad, with a 12-week duration. The second card is 'Animal Waste Management' by Dr. Vipin Maurya, Dr. Utkarsh Kumar Tripathi, Dr. ... at Institute of Agricultural Sciences, BHU, with a 4-week duration. The third card is 'Applied and Economic Zoology' by Dr. Dalie Dominic A at St. Mary's College, Thrissur, with a 12-week duration. To the left of the course cards is a 'FILTERS' sidebar with dropdown menus for 'National Coordinator', 'Course Mode', 'Course Duration', 'Course Language', 'Course Exam Date', and 'Course Credits'. At the bottom right of the page, there is a small cartoon robot character.

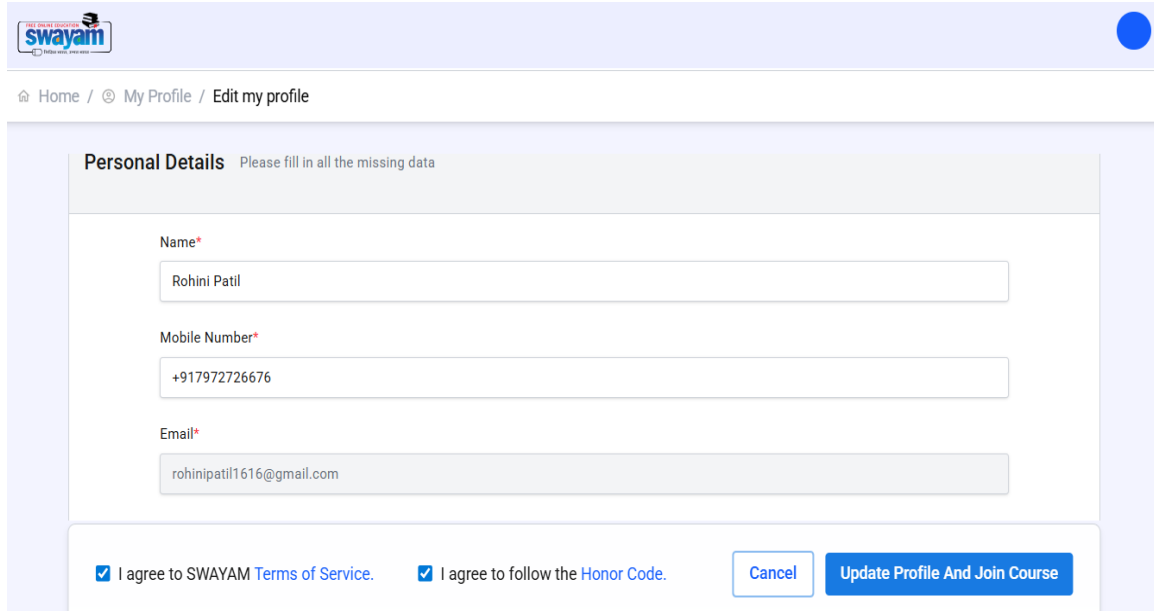
Step 3: Searched for the Course

I used the search bar to find the course titled “Animal Biotechnology.”

The screenshot shows the Swayam course page for 'Animal Biotechnology'. The page has a light blue header with the Swayam logo and a search bar. Below the header, there is a 'Course Information' section. On the left side, there is a sidebar with a 'Welcome to the course' message, the course title 'Animal Biotechnology', the instructor 'By Dr. Jayasree.S', the institution 'Mercy College, Palakkad', and the number of learners enrolled '2760'. The sidebar also has a list of links: 'Course Information', 'Summary', 'Course outline', 'Books and References', 'Instructor Bio', and 'Course Certificate'. The main content area features a large video player with a red play button and the text 'ANIMAL BIOTECHNOLOGY'. Above the video player, there is a 'MOOC ANIMAL BIOTECHNOLOGY PROMO' banner. To the right of the video player, there are buttons for 'Watch later' and 'Share'. At the bottom right of the page, there is a blue button labeled 'Join the Course'.

Step 4: Registered / Logged In

I created a new account into the SWAYAM platform using my email ID and password.



swayam

Home / My Profile / Edit my profile

Personal Details Please fill in all the missing data

Name*

Rohini Patil

Mobile Number*

+917972726676

Email*

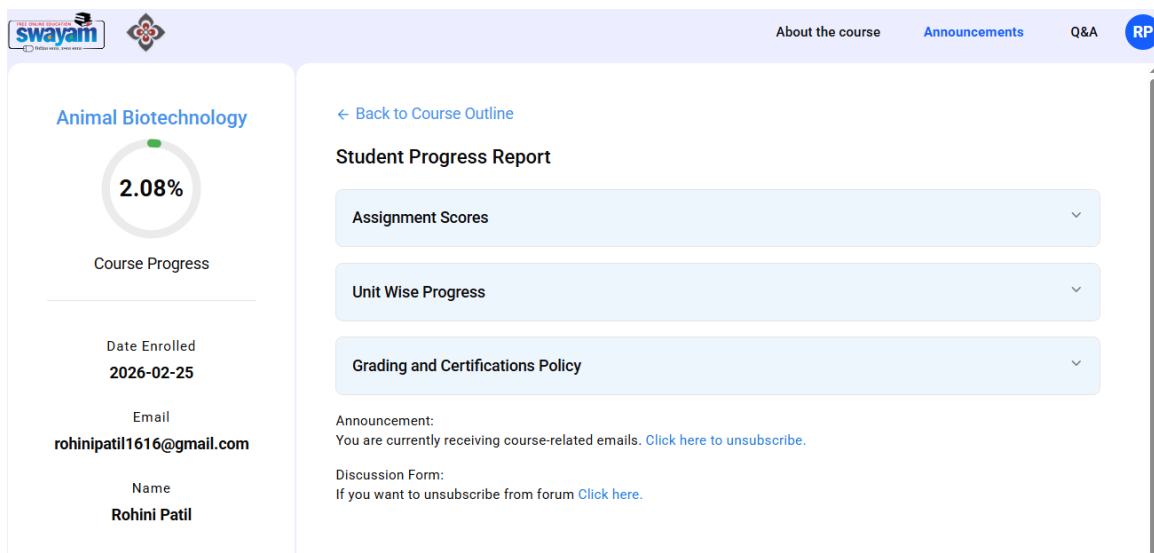
rohinipatil1616@gmail.com

☒ I agree to SWAYAM [Terms of Service](#). ☒ I agree to follow the [Honor Code](#).

[Cancel](#) [Update Profile And Join Course](#)

Step 5: Enrolled in the Course

After logging in, I **Enrolled** in the course.



swayam

About the course Announcements Q&A RP

Animal Biotechnology

2.08%

Course Progress

Date Enrolled
2026-02-25

Email
rohinipatil1616@gmail.com

Name
Rohini Patil

[← Back to Course Outline](#)

Student Progress Report

Assignment Scores

Unit Wise Progress

Grading and Certifications Policy

Announcement:
You are currently receiving course-related emails. [Click here to unsubscribe](#).

Discussion Form:
If you want to unsubscribe from forum [Click here](#).

ACTIVITY 5

IKS Integrated Video / Presentation:

Introduction:

The Indian Knowledge System (IKS) refers to the vast and profound body of knowledge developed in India over thousands of years across various disciplines including science, mathematics, philosophy, medicine, art, and education. It reflects India's intellectual traditions, cultural practices, and scientific advancements that have significantly contributed to global knowledge.

In alignment with the National Education Policy (NEP 2020), there is a renewed emphasis on integrating indigenous knowledge systems into mainstream education. NEP 2020 recognizes the importance of connecting learners with India's rich heritage while ensuring global competence. The integration of IKS aims to promote holistic development by nurturing cognitive skills, ethical values, creativity, and cultural awareness.

Thus, incorporating IKS into classroom teaching not only enhances subject understanding but also instills pride in India's intellectual legacy.

1. Concept of Indian Knowledge System (IKS)

The Indian Knowledge System encompasses diverse domains of traditional knowledge that evolved through systematic observation, experimentation, and philosophical inquiry. It includes:

- The Vedas and Upanishads (spiritual and philosophical foundations)
- Atharva Veda (source of various mathematical sutras)
- Ayurveda (ancient medical science)
- Vedic Mathematics
- Ancient Indian astronomy, metallurgy, architecture, linguistics, and logic

IKS is not merely historical knowledge; it is a dynamic and holistic framework that integrates theory with practical application. It emphasizes harmony between mind, body, and environment. Unlike fragmented learning models, IKS promotes interconnected understanding and lifelong learning.

The core philosophy of IKS focuses on:

- Experiential learning
- Logical reasoning
- Ethical conduct
- Self-discipline
- Intellectual independence

2. Usefulness of IKS in Teaching

The integration of IKS into teaching practices provides multiple pedagogical benefits:

- **Enhances Conceptual Understanding**

Traditional Indian methods often simplify complex concepts through logical patterns and mental techniques. This improves clarity and long-term retention.

- **Promotes Experiential and Activity-Based Learning**

IKS-based teaching encourages demonstrations, discussions, and practical applications, making learning more engaging.

- **Strengthens Cultural Identity**

Students become aware of India's scientific and intellectual contributions, which fosters confidence and national pride.

- **Develops Higher-Order Thinking Skills**

Many traditional methods require analytical reasoning and pattern recognition, thereby enhancing critical thinking.

- **Reduces Academic Anxiety**

In subjects like Mathematics, simplified techniques reduce fear and increase self-confidence among learners.

Thus, IKS transforms teaching from rote memorization to meaningful understanding.

3. Selected School Subject

Grade: 7

Subject: Mathematics

Topic: Vedic Mathematics

The selected IKS component for integration is Vedic Mathematics, particularly techniques for two-digit multiplication.

Mathematics at the middle school level often creates anxiety among students. Introducing Vedic methods at this stage helps learners develop strong foundational skills, improve calculation speed, and build confidence in problem-solving.

This topic allows a comparative approach between:

- ✓ Traditional long multiplication method**
- ✓ Vedic multiplication techniques**

Such comparison encourages analytical thinking and deeper conceptual clarity.

4. Integration of IKS with the Subject

Vedic Mathematics is based on 16 Sutras (aphorisms) and 13 Upa-Sutras (sub-aphorisms) derived from ancient Indian texts. These Sutras provide short, logical formulas for solving arithmetic problems mentally and efficiently.

One important multiplication technique is:

Vertically and Crosswise Method (Urdhva Tiryagbhyam)

This method simplifies multiplication by breaking it into vertical and crosswise steps.

Example: Multiply 23×14

Step 1: Multiply last digits

$3 \times 4 = 12 \rightarrow$ Write 2, carry 1

Step 2: Cross multiply and add

$(2 \times 4) + (3 \times 1) = 8 + 3 = 11$

Add carry $\rightarrow 12 \rightarrow$ Write 2, carry 1

Step 3: Multiply first digits

$2 \times 1 = 2$

Add carry $\rightarrow 3$

Final Answer = 322

Educational Impact of Integration:

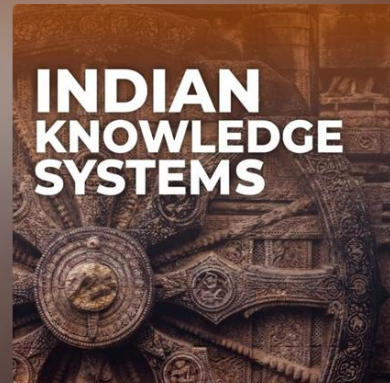
- ✓ Encourages mental arithmetic
- ✓ Reduces lengthy written steps
- ✓ Enhances speed and accuracy
- ✓ Builds logical sequencing skills
- ✓ Makes mathematics enjoyable

By integrating Vedic Mathematics into Grade 7 curriculum, students not only learn a faster technique but also understand India's historical contribution to mathematical sciences.

5.OUTLINE OF PPT:

Introduction to Indian Knowledge System (IKS)

Prepared By : Rohini Patil. (Roll No. 4)



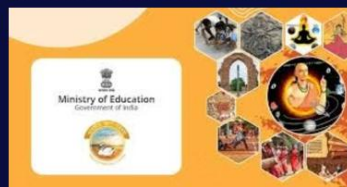
Concept of Indian Knowledge System

The Indian Knowledge System (IKS) encompasses the rich, traditional intellectual heritage of India. Guided by the **National Education Policy (NEP 2020)**, there's a growing imperative to seamlessly integrate these profound traditions into contemporary education.

IKS represents a comprehensive exploration of India's vibrant culture, time-honored traditions, enduring values, and deep-rooted knowledge systems. It is a holistic framework designed to foster the all-round development of students.

It draws wisdom from:

- Vedas and Upanishads
- Ayurveda (traditional medicine)
- Vedic Mathematics
- Ancient Indian Sciences (astronomy, metallurgy, linguistics, etc.)



Objectives of Integrating IKS in Math



Introduce Vedic Mathematics

Familiarize students with this ancient system as an alternative approach to arithmetic.



Increase Calculation Speed

Develop techniques for faster and more efficient mathematical problem-solving.



Develop Mental Math Skills

Cultivate the ability to perform calculations mentally, reducing reliance on external tools.



Promote Cultural Awareness

Educate students about India's rich contributions to global knowledge, fostering pride and appreciation.

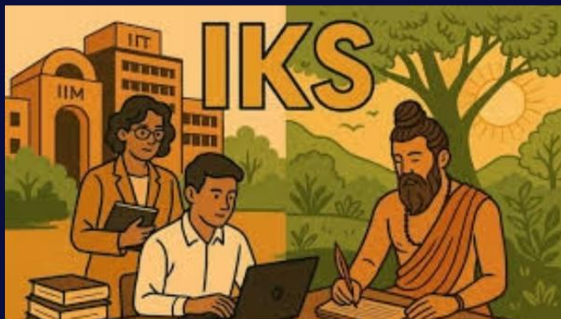


Enhance Confidence

Boost student self-assurance in tackling mathematical challenges through varied approaches.

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Selected Teaching Component:



Context for Integration

- **Grade:** 7
- **Subject:** Mathematics
- **Topic:** Vedic Mathematics

The chosen IKS component for integration is **Vedic Mathematics**. This specific topic allows for a clear demonstration of Vedic principles' efficiency compared to traditional methods.

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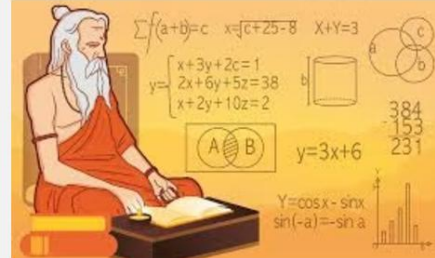
Vedic Mathematics: The IKS Component

Vedic Mathematics is an ancient Indian system that offers a unique approach to mathematical calculations. It is based on 16 **Sutras** (aphorisms) and 13 **Upa-Sutras** (sub-aphorisms) derived from the Atharva Veda.

This system is renowned for its ability to:

- Enable faster and simpler calculations
- Provide elegant and creative solutions
- Reduce mathematical anxiety

For two-digit multiplication, we will focus on the **Vertically and Crosswise Method (Urdhva Tiryagbhyam)** or the **Nikhilam Sutra**.



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Vedic Method vs. Modern Method: An Example

Traditional Long Multiplication

$$\begin{array}{r} 23 \\ \times 14 \\ \hline 92 \quad (23 \times 4) \\ 230 \quad (23 \times 10) \\ \hline 322 \end{array}$$

- Step-by-step written multiplication
- Requires more writing space
- Focuses on sequential operations

Vedic Mathematics: Vertically and Crosswise

Example: Multiply **23 × 14**

1. **Step 1:** Multiply last digits: $3 \times 4 = 12$ → Write 2, carry 1
2. **Step 2:** Cross multiply and add: $(2 \times 4) + (3 \times 1) = 8 + 3 = 11$. Add carry → $11 + 1 = 12$ → Write 2, carry 1
3. **Step 3:** Multiply first digits: $2 \times 1 = 2$. Add carry → $2 + 1 = 3$

Final Answer = 322

- Faster calculation
- Encourages mental arithmetic
- Reduces lengthy written steps

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Teaching Methodologies

Question-Answer Method

Engaging students through interactive queries to stimulate critical thinking and recall prior knowledge.

Demonstration Method

Teacher models Vedic multiplication steps clearly on the board, showing both traditional and Vedic approaches.

Group Work

Collaborative problem-solving in small groups to foster peer learning and discussion.

Practice Method

Ample opportunities for students to apply new Vedic techniques through exercises and worksheets.

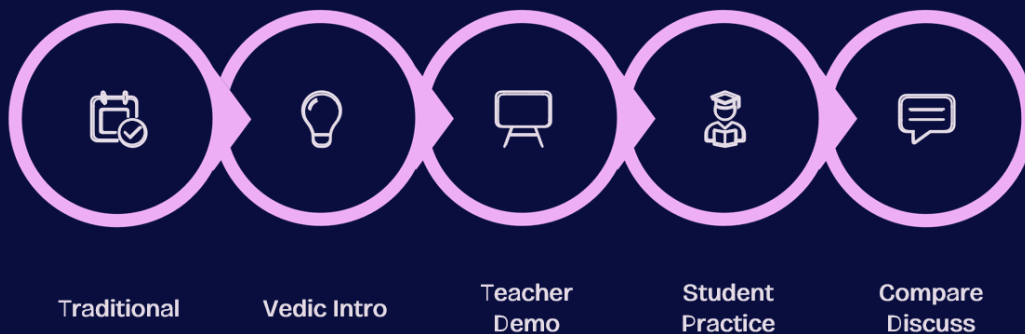
Comparative Method

Directly contrasting Vedic and modern methods to highlight the advantages and efficiency of the former.

Our approach emphasizes student-centered and activity-based learning, making mathematics engaging and accessible.

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The Teaching Process



This structured approach ensures students first understand the traditional method, then are progressively introduced to, and proficient in, the Vedic approach, culminating in a critical comparative analysis.

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Educational Tools & Evaluation

Educational Tools

- **Blackboard and Chalk**
For interactive demonstrations and step-by-step problem-solving.
- **PPT Presentation**
Visual aids for concepts, examples, and historical context of IKS.
- **Worksheets**
Structured practice for reinforcing Vedic Sutras and building proficiency.
- **Flashcards**
For quick recall of basic facts and mental math exercises.
- **Digital Videos**
Engaging supplementary content showcasing Vedic Math techniques and applications.

Evaluation Methods

- **Oral Questioning**
Assessing understanding and immediate application of Vedic principles.
- **Practice Examples**
Monitoring progress and identifying areas for further support.
- **Group Competition**
Encouraging friendly rivalry and motivating rapid calculation skills.
- **Worksheets**
Formal assessment of accuracy and understanding of different Sutras.
- **Quick Mental Math Test**
Evaluating improvement in speed and confidence in mental calculations.

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Conclusion & References

Key Takeaways

Integrating Vedic Mathematics transforms the learning experience, making mathematics more accessible, engaging, and enjoyable. It equips students with powerful tools for faster calculations and instills greater confidence.

By embracing the Indian Knowledge System, we not only enhance mathematical proficiency but also cultivate a deeper appreciation for India's rich cultural heritage and promote holistic learner development, aligning with the vision of NEP 2020.

References

- Mathematics Textbook – Grade 7
- Vedic Mathematics by Jagadguru Bharati Krishna Tirthaji
- National Education Policy (NEP 2020) – Indian Knowledge System
- Relevant Educational Websites

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Thank You!

9. Reflection

In this project, various components of ICT and digital education were studied in depth, and their educational usefulness was clearly explained.

In the introductory section, the role and necessity of ICT in the modern education system were highlighted. In today's technology-driven era, ICT has become essential for making education more effective, engaging, and student-centered.

While explaining the importance of ICT and digital education, it became clear how e-learning, smart classrooms, online tools, and multimedia are used in the teaching process. Digital education makes learning easier to understand, time-saving, labor-saving, and interactive.

While studying the need and advantages of ICT and digital education, many positive outcomes were observed, such as easy access to knowledge, opportunities for self-learning, access to global information, skill development, and technological literacy.

Through the activity of creating a digital newsletter, various educational websites were studied. Their concepts, features, classroom usefulness, and presentation structure were understood. This helped students gain information about reliable educational resources and develop digital presentation skills.

While creating a digital assessment tool (online quiz), practical experience was gained regarding the concept of digital assessment, question paper design, answer keys, feedback systems, and classroom usefulness. Digital assessment makes it easier to obtain immediate results, maintain transparency, and analyze students' progress.

Through the digital safety awareness poster/video, the concept of cyber safety was explained. By delivering key messages on the topic of online safety, an effort was made to develop safe digital habits. This helps in shaping responsible digital citizens.

Through studying MOOCs, I understood that digital learning is transforming the traditional education system. MOOCs provide:

- Flexibility in time and place of learning

- Access to quality content from expert educators
- Self-paced learning opportunities
- Certification that enhances professional growth

Learning about IKS helped me understand the depth and richness of Indian heritage. It made me realize that:

- Ancient Indian education systems like the Gurukul system focused on holistic development.
- Indian mathematics, astronomy, and medicine were highly advanced.
- Values such as discipline, ethics, and harmony with nature were emphasized.

As a future teacher, I feel that integrating IKS in modern classrooms will:

- Develop cultural awareness among students
- Build moral and ethical values
- Promote national pride and identity
- Encourage sustainable living practices

10) Reference

1. NCERT. (2020). *ICT in Education – Curriculum and Guidelines*. New Delhi.
2. CBSE. (2019). *Digital Learning and Assessment Framework*.
3. Ministry of Education, Government of India. (2020). *National Education Policy (NEP 2020)*.
4. SWAYAM Portal – www.swayam.gov.in
5. DIKSHA Platform – www.diksha.gov.in
6. National Cyber Crime Reporting Portal – www.cybercrime.gov.in
7. Google Safety Center – safety.google
8. Microsoft Education Resources – education.microsoft.com
9. Coursera – www.coursera.org
10. edX – www.edx.org
11. NPTEL – www.nptel.ac.in
12. Ministry of Education. (2022). *Indian Knowledge System (IKS) Initiative*.
13. UNESCO. (2021). *Digital Learning Guidelines*.